

ASSESSMENT TOOL – 1

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO3	Assessment	Experiments
Weightage		Total Marks	

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Section / Batch	ECE-2024	Date of Submission	
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks	20	Obtained Marks	20
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Code of Conduct:

(Handwritten signature and date 10/8/24)

I Cindy R / 192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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Experiment 1: TCP Three-Way Handshake

Aim

To capture packets during TCP connection establishment using Wireshark and identify the SYN, SYN-ACK, and ACK packets.

Tools Required

- Wireshark
- Internet connection
- Web browser

Procedure

1. Open Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Open a website (e.g., www.google.com).
5. Stop the capture.
6. Apply the filter:

tcp

7. Identify the TCP three-way handshake.

Observation

Packet	Source	Destination	TCP Flag	Purpose
1	Client	Server	SYN	Requests connection
2	Server	Client	SYN, ACK	Acknowledges SYN and accepts request
3	Client	Server	ACK	Confirms connection establishment

Explanation of TCP Flags

SYN

- Synchronizes sequence numbers.
- Initiates the TCP connection.

SYN-ACK

- Server acknowledges the SYN.
- Sends its own SYN to synchronize.

ACK

- Client acknowledges server's SYN.
- Connection becomes established.

Result Analysis

TCP establishes a reliable connection using the three-way handshake. Sequence numbers are synchronized before data transfer begins, ensuring reliable communication.

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No.	Time	Source	Destination	Length	Info
1	0.000000	192.168.1.100	192.168.1.1	60	SYN 33956 → 80 [RST] Seq=33956 Window=0 Len=0
2	0.000000	192.168.1.1	192.168.1.100	60	ACK 80 → 33956 [RST] Seq=33956 Window=0 Len=0
3	0.000000	192.168.1.1	192.168.1.100	60	ACK 33956 → 80 [RST] Seq=33956 Window=0 Len=0
4	0.000000	192.168.1.100	192.168.1.1	60	ACK 80 → 33956 [RST] Seq=33956 Window=0 Len=0
5	0.000000	192.168.1.100	192.168.1.1	60	ACK 80 → 33956 [RST] Seq=33956 Window=0 Len=0
6	0.000000	192.168.1.100	192.168.1.1	60	ACK 80 → 33956 [RST] Seq=33956 Window=0 Len=0
7	0.000000	192.168.1.100	192.168.1.1	60	ACK 80 → 33956 [RST] Seq=33956 Window=0 Len=0
8	0.000000	192.168.1.100	192.168.1.1	60	ACK 80 → 33956 [RST] Seq=33956 Window=0 Len=0

Experiment 2: DNS Query Using UDP

Aim

Capture a DNS query and compare UDP and TCP headers.

Procedure

1. Clear DNS cache (optional).
2. Start Wireshark.
3. Apply filter:

`dns`

or

`udp.port==53`

4. Open a website.
5. Stop capture.

Observation

Typical DNS Packet:

Client → DNS Server

Standard Query

DNS Server → Client

Standard Query Response

Header Comparison

Feature	UDP	TCP
Header Size	8 Bytes	20 Bytes (minimum)
Sequence Number	No	Yes
Acknowledgment	No	Yes
Connection	Connectionless	Connection-oriented
Reliability	No	Yes
Speed	Faster	Slightly slower

Analysis

UDP has a much smaller header than TCP.

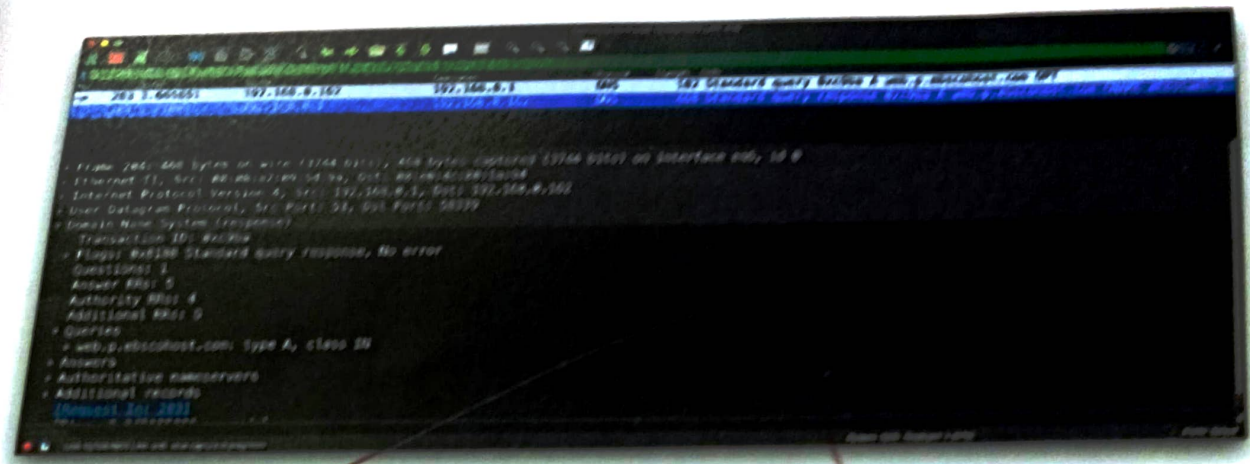
Since UDP does not include:

- Sequence numbers

- Acknowledgments
- Flow control

its overhead is lower, making it faster.

However, lost packets are not retransmitted.



Experiment 3: Packet Loss – TCP vs UDP

Aim

Observe TCP retransmission during packet loss and compare it with UDP.

Procedure

1. Start Wireshark.
2. Begin downloading a file.
3. Disconnect Wi-Fi for a few seconds.
4. Reconnect.
5. Stop capture.
6. Apply filters:

TCP:

tcp.analysis.retransmission

UDP:

udp

Observation

TCP Capture:

TCP Retransmission

Multiple retransmitted packets appear until acknowledgment is received.

UDP Capture:

Packets are transmitted once.

No retransmission occurs.

Comparison

Feature	TCP	UDP
Packet Loss Recovery	Yes	No
Retransmission	Yes	No
ACK Required	Yes	No
Reliability	High	Low
Delay	Higher	Lower

Analysis

TCP detects packet loss using acknowledgments and timers.

If acknowledgments are not received, packets are retransmitted.

UDP simply sends packets without checking delivery.

Hence,

TCP

- Reliable
- Slower

UDP

- Faster
- Less reliable

No.	Time	Source	Destination	Protocol	Length	Info
318	34.139720	142.250.183.238	192.168.1.18	QUIC	67	Protected Payload (KPB)
319	34.139970	192.168.1.18	142.250.183.238	QUIC	74	Protected Payload (KPB), DCID=ec5c8aa1740ae9cb
320	34.140222	142.250.183.238	192.168.1.18	QUIC	719	Protected Payload (KPB)
321	34.140422	142.250.183.238	192.168.1.18	QUIC	202	Protected Payload (KPB)
322	34.140688	192.168.1.18	142.250.183.238	QUIC	77	Protected Payload (KPB), DCID=ec5c8aa1740ae9cb
323	34.140963	142.250.183.238	192.168.1.18	QUIC	719	Protected Payload (KPB)
324	34.140892	192.168.1.18	142.250.183.238	QUIC	75	Protected Payload (KPB), DCID=ec5c8aa1740ae9cb
325	34.141004	142.250.183.238	192.168.1.18	QUIC	66	Protected Payload (KPB)
326	35.065232	192.168.1.18	20.42.73.26	TCP	60	63324 → 443 [SYN] Seq=0 Win=0 Len=0 SACK_PERM
327	35.226116	20.42.73.26	192.168.1.18	TCP	60	443 → 63324 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0 SACK_PERM
328	35.226257	192.168.1.18	20.42.73.26	TCP	54	63324 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
329	35.227669	192.168.1.18	20.42.73.26	TLSv1.3	439	Client Hello
330	35.531305	20.42.73.26	192.168.1.18	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
331	35.531497	192.168.1.18	20.42.73.26	TCP	54	63324 → 443 [ACK] Seq=300 Ack=100 Win=261888 Len=0
332	35.530614	192.168.1.18	20.42.73.26	TLSv1.3	530	Change Cipher Spec, Client Hello
333	35.845853	20.42.73.26	192.168.1.18	TLSv1.3	1406	Server Hello
334	35.845963	192.168.1.18	20.42.73.26	TCP	54	63324 → 443 [ACK] Seq=842 Ack=1512 Win=262144 Len=0
335	35.846313	20.42.73.26	192.168.1.18	TCP	1466	443 → 63324 [ACK] Seq=1512 Ack=842 Win=4196064 Len=1412 [TCP segment of a reassembled PDU]

Experiment 4: DNS Resolution Analysis

Aim

Capture DNS traffic while resolving a domain name.

Procedure

1. Open Wireshark.
2. Start capture.
3. Apply filter:

dns

4. Visit:

www.google.com

5. Stop capture.

Observation

Example Query:

Query Name:

www.google.com

Query Type

Common DNS Query Types

Type	Purpose
A	IPv4 Address
AAAA	IPv6 Address
MX	Mail Server
NS	Name Server
CNAME	Canonical Name

Example:

Type:

A

Response:

142.250.xxx.xxx

DNS Resolution Time

Example:

Request Time:

2.051 s

Response Time:

2.064 s

Resolution Time

2.064 - 2.051

= 0.013 seconds

= 13 ms

Why DNS Uses UDP

- Small packet size
- No connection setup
- Low latency
- Fast query-response mechanism
- Minimal overhead

TCP is used only when:

- Zone transfer occurs
- DNS response exceeds UDP size limits
- DNSSEC requires larger responses

```
> Ethernet II, Src: Cisco_a5:8d:69 (70:01:b5:a5:8d:69), Dst: Dell_91:ff:73 (70:b5:e8:91:ff:73)
> Internet Protocol Version 4, Src: 10.143.70.254, Dst: 10.143.90.167
> User Datagram Protocol, Src Port: 53, Dst Port: 50302
> Domain Name System (response)
  Transaction ID: 0x7b13
  Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 7
  Authority RRs: 0
  Additional RRs: 0
  Queries
  Answers
  [Unsolicited: True]
0020  5a a7 00 35 c4 7e 00 a4 61 02 7b 13 81 80 00 01  Z 5 . . . a { . . .
0030  00 07 00 00 00 00 07 68 69 73 74 6f 72 79 06 67  . . . h istory-g
0040  6f 6f 67 6c 65 03 63 6f 6d 00 00 01 00 01 c0 0c  oogle-co m . . .
0050  00 05 00 01 00 00 00 08 00 0c 07 68 69 73 74 6f  . . . . . histo
0060  72 79 01 6c c0 14 c0 30 00 01 00 01 00 00 01 1f  ry-1 . . .
0070  00 04 4a 7d 44 8a c0 30 00 01 00 01 00 00 01 1f  .J)D . . .
0080  00 04 4a 7d 44 64 c0 30 00 01 00 01 00 00 01 1f  .J)Dd . . .
0090  00 04 4a 7d 44 65 c0 30 00 01 00 01 00 00 01 1f  .J)De . . .
00a0  00 04 4a 7d 44 71 c0 30 00 01 00 01 00 00 01 1f  .J)Dq . . .
```

Overall Result

1. TCP establishes reliable communication using the **three-way handshake (SYN, SYN-ACK, ACK)**.
2. UDP has a **smaller 8-byte header** and no sequence numbers or acknowledgments, making it **faster** but less reliable.
3. During packet loss, **TCP retransmits lost packets**, whereas **UDP does not**, demonstrating TCP's reliability at the cost of additional delay.
4. DNS queries are typically sent over **UDP** because its lightweight, connectionless design enables **fast domain name resolution** with minimal overhead.

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